
MA 578: BAYESIAN STATISTICS

Homework 6

Due: November 25 at 12:30 PM (start of class)

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1. The files `school1.dat` through `school8.dat` give weekly hours spent on homework for students sampled from eight different schools. Obtain posterior distributions for the true means for the eight different schools using a hierarchical normal model with the following prior parameters:

$$\mu_0 = 7, \gamma_0^2 = 5, \tau_0^2 = 10, \eta_0 = 2, \sigma_0^2 = 15, \nu_0 = 2.$$

- (a) Run a Gibbs sampling algorithm to approximate the posterior distribution of $\{\theta, \sigma^2, \mu, \tau^2\}$. Assess the convergence of the Markov chain (using trace plot and auto-correlation plot), and find the effective sample size for $\{\sigma^2, \mu, \tau^2\}$. Run the chain long enough so that the effective sample sizes are all above 1,000.
 - (b) Compute posterior means and 95% confidence regions for $\{\sigma^2, \mu, \tau^2\}$. Also, compare the posterior densities to the prior densities visually, and discuss what was learned from the data.
 - (c) Plot the posterior density of $R = \frac{\tau^2}{\sigma^2 + \tau^2}$ and compare it to a plot of the prior density of R . Describe the evidence for between-school variation.
 - (d) Obtain the posterior probability that θ_7 is smaller than θ_6 , as well as the posterior probability that θ_7 is the smallest of all the θ 's.
 - (e) Plot the sample averages $\bar{y}_1, \dots, \bar{y}_8$ against the posterior expectations of $\theta_1, \dots, \theta_8$, and describe the relationship. Also compute the sample mean of all observations and compare it to the posterior mean of μ .
2. Researchers interested in identifying the optimal planting density for a type of perennial grass performed the following randomized experiment: Ten different plots of land were each divided into eight subplots, and planting densities of 2, 4, 6 and 8 plants per square meter were randomly assigned to the subplots, so that there are two subplots at each density in each plot. At the end of the growing season the amount of plant matter yield was recorded in metric tons per hectare. These data appear in the file `pdensity.dat`. The researchers want to fit a model like $y = \beta_1 + \beta_2 x + \beta_3 x^2 + \epsilon$, where y is yield and x is planting density, but worry that since soil conditions vary across plots they should allow for some across-plot heterogeneity in this relationship. To accommodate this possibility we will analyze these data using the hierarchical linear model.
 - (a) Before we do a Bayesian analysis we will get some ad hoc estimates of these parameters via least squares regression. Fit the model $y = \beta_1 + \beta_2 x + \beta_3 x^2 + \epsilon$ using OLS for each group, and make a plot showing the heterogeneity of the least squares regression lines. From the least squares coefficients find ad hoc estimates of θ and Σ . Also obtain an estimate of σ^2 by combining the information from the residuals across the groups.
 - (b) Now we will perform a Bayesian analysis of the data using the following distributions as prior distributions:

$$\Sigma^{-1} \sim \text{Wishart}(4, \hat{\Sigma}^{-1})$$

$$\theta \sim \mathcal{N}(\hat{\theta}, \hat{\Sigma})$$

$$\sigma^2 \sim \text{Inv-Gamma}(1, \hat{\sigma}^2)$$

where, $\hat{\Sigma}^{-1}, \hat{\theta}, \hat{\sigma}^2$ are the estimates you obtained in (a). Use a Gibbs sampler to approximate posterior expectations of β for each group j , and plot the resulting regression lines. Compare to the regression lines in (a) above and describe why you see any differences between the two sets of regression lines.

- (c) From your posterior samples, plot marginal posterior and prior densities of θ and the elements of Σ . Discuss the evidence that the slopes or intercepts vary across groups.
- (d) Suppose we want to identify the planting density that maximizes average yield over a random sample of plots. Find the value x_{max} of x that maximizes expected yield, and provide a 95% posterior predictive interval for the yield of a randomly sampled plot having planting density x_{max} .